Opinion Pooling IV: Linear Pooling

2026-02-11

Introducing Linear Pooling

Recap: Where We Left Off

Last time we discussed:

- Formal axioms for pooling strategies; and
- Which combinations of them are impossible.

Today: The most intuitive pooling strategy—**linear pooling**—and why commutativity matters so much.

What Is Linear Pooling?

Linear pooling is weighted averaging of credences.

- Take each person's credence in proposition A.
- Multiply each by a weight (reflecting their importance/expertise).
- Add them up.

Formally: $Pr_G(A) = w_1 Pr_1(A) + w_2 Pr_2(A) + ... + w_n Pr_n(A)$

where the weights satisfy: $w_i \geq 0$ and $\sum_i w_i = 1$.

A Simple Example

Three epidemiologists estimate the probability of polio eradication by 2030:

- Anya: 0.20
- Bon: 0.60
- Carys: 0.70

Equal weighting (each gets $w = 1/3$):

$$Pr_G = \frac{1}{3}(0.20) + \frac{1}{3}(0.60) + \frac{1}{3}(0.70) = 0.50$$

Unequal Weights

Suppose Bon is the senior researcher, so we give her more weight:

- $w_1 = 0.2$ (Anya)
- $w_2 = 0.5$ (Bon)
- $w_3 = 0.3$ (Carys)

Then: $Pr_G = 0.2(0.20) + 0.5(0.60) + 0.3(0.70) = 0.55$

The group credence shifts toward the more heavily weighted expert.

Why Linear Pooling?

It's the most natural aggregation method:

- Really easy to understand, and to use. It's what you'd use by default.
- Respects the idea of "compromise"—everyone's opinion counts.
- Can flexibly accommodate different weights for different people.

It also turns out to have some nice formal properties.

Properties of Linear Pooling

What Linear Pooling Satisfies

Linear pooling does well on several of our axioms from last time:

- **Unanimity**: If everyone agrees, so does the group.
- **Non-dictatorship**: No single person determines the outcome (unless you choose dictatorial weights).
- **Bounded pooling**: The group credence stays between the minimum and maximum individual credences.

These all follow straightforwardly from the weighted average structure.

Unanimity

If $Pr_i(A) = x$ for all i, then:

$$Pr_G(A) = w_1 \cdot x + w_2 \cdot x + ... + w_n \cdot x$$

$$= x(w_1 + w_2 + ... + w_n)$$

$$= x \cdot 1 = x$$

The weights sum to 1, so we get back the unanimous value.

Suppose the minimum credence is m and the maximum is M.

Since each $Pr_i(A)$ is between m and M:

$$Pr_G(A) = \sum_i w_i Pr_i(A)$$

Because the weights are non-negative and sum to 1, this is a **convex combination**.

Convex combinations always lie within the range of the values being combined.

So: $m \leq Pr_G(A) \leq M$.

Eventwise Independence

Linear pooling also satisfies **eventwise independence**.

Recall: The group credence in A should depend only on individual credences in A, not on credences about other propositions.

For linear pooling: $Pr_G(A) = \sum_i w_i Pr_i(A)$

This formula only uses $Pr_1(A), Pr_2(A), ..., Pr_n(A)$.

It doesn't matter what anyone thinks about B, C, or anything else.

The Characterization Theorem

A Remarkable Result

We've just seen that linear pooling satisfies unanimity and eventwise independence.

Here's the striking fact: These two properties **uniquely characterize** linear pooling.

Theorem (Aczél & Wagner 1980; McConway 1981):

A pooling function satisfies both unanimity preservation and eventwise independence **if and only if** it is linear pooling.

This raises the stakes for our next question...

The Impossibility Preview

We've established:

- Linear pooling $\leftrightarrow$ Unanimity + Eventwise Independence

But we also want:

- External Bayesianity (commutativity with updating)

The problem: Can we have all three?

No. Let's work through an example to see why.

The Problem: Commutativity

Groups don't just form opinions once. They also **update** those opinions in light of new evidence.

This raises a procedural question: What's the right order of operations?

Two Procedures

Procedure 1: Pool first, then update

1. Aggregate individual credences into group credence.
2. New evidence E arrives.
3. Update the group credence on E using Bayes' rule.

Procedure 2: Update first, then pool

1. New evidence E arrives.
2. Each individual updates their credence on E using Bayes' rule.
3. Aggregate the updated individual credences.

Should the Order Matter?

Intuitively, we might think these should give the same result.

- After all, everyone has the same evidence.
- Everyone is using the same updating rule (Bayes' rule).
- We're using the same pooling function in both cases.

This is the property called **commutativity** or **external Bayesianity**.

External Bayesianity (Formal)

A pooling function satisfies **external Bayesianity** if:

$$F(Pr_1(\cdot|E), ..., Pr_n(\cdot|E)) = F(Pr_1, ..., Pr_n)(\cdot|E)$$

for all evidence E.

Left side: Update first, then pool.

Right side: Pool first, then update.

Question: Does linear pooling satisfy this?

Let's check with a simple example.

There are two people, call them 1 and 2. And there are two propositions, A and B.

Both of them agree that A and B are independent.

Person 1

Both A and B are really likely.

- $Pr_1(A) = Pr_1(B) = 0.9$, so by the independence assumption
- $Pr_1(A \wedge B) = 0.81$, and by the rules for negation
- $Pr_1(\neg A) = Pr_1(\neg B) = 0.1$, then applying independence a few times,
- $Pr_1(A \wedge \neg B) = Pr_1(\neg A \wedge B) = 0.09$
- $Pr_1(\neg A \wedge \neg B) = 0.01$

Person 2

Both A and B are really unlikely.

- $Pr_2(A) = Pr_2(B) = 0.1$, so by the independence assumption
- $Pr_2(A \wedge B) = 0.01$, and by the rules for negation
- $Pr_2(\neg A) = Pr_2(\neg B) = 0.9$, then applying independence a few times,
- $Pr_2(A \wedge \neg B) = Pr_2(\neg A \wedge B) = 0.09$
- $Pr_2(\neg A \wedge \neg B) = 0.81$

Equal Weight Pooling

So pre-update for the group we get

- $Pr_G(A \wedge B) = 0.41$, because that's (0.81 + 0.01)/2,
- $Pr_G(\neg A \wedge \neg B) = 0.41$, because that's (0.01 + 0.81)/2,
- $Pr_G(A \wedge \neg B) = Pr_G(\neg A \wedge B) = 0.09$, because they agree on this.

Putting this into the rules for conditional probability, we get:

- $Pr_G(A|B) = 0.82$, because $Pr_G(A \wedge B) = 0.41$, and $Pr_G(B) = 0.5$.

Updating

Let's say that they are going to run an experiment on whether B is true. Neither of them actually care, because they both think it's independent, but they've been told to run it so they will.

Now we have two options for pooling.

1. Pool-then-update.
2. Update-then-pool.

Do they lead to the same result?

Let's assume that the result of the experiment is B is true; it will go the same way when B is false.

Pool-Then-Update

First, we work out Pr_G pre-update, as above.

Then the post-update, $Pr_G(A)$ is the value of $Pr_G(A|B)$ pre-update.

That is, the post-update value of $Pr_G(A) = 0.82$.

Update-Then-Pool

Both of 1 and 2 think A and B are independent. So they don't change their mind about A when learning B.

So post-update we have:

- $Pr_1(A) = 0.9$, just like pre-update, and
- $Pr_2(A) = 0.1$, also just like pre-update.

And then post-update $Pr_G(A) = 0.5$. Which is a very different number to 0.82.

That is a very unrealistic example, but the general pattern obtains.

In realistic cases, the two procedures differ by a small amount, but it turns out they almost always differ.

Linear pooling violates external Bayesianity.

Why This Matters

If the order matters, then the group's final opinion depends on **meeting logistics**.

- Did you convene the committee before or after the data came in?
- This is an arbitrary, non-epistemic factor.
- Yet it changes the group's beliefs about the world.

That seems weird.

The Manipulation Problem

Worse: Someone who controls the process can manipulate the outcome.

Suppose you're chairing a committee and you prefer Procedure 1's result.

- You can ensure that result by convening the committee before the evidence arrives.
- If someone else chairs and prefers Procedure 2's result, they schedule after the evidence.
- The group's "rational" belief depends on procedural maneuvering.

A Sure Loss Problem

The problem goes beyond arbitrariness. A group that uses update-then-pool can be **Dutch booked**.

A bookie can offer the group a package of bets, each of which is fair by the group's own credences at the time it is offered, but which **guarantee a net loss** for the group no matter what happens.

Let's see how this works with our example. The key fact is the gap between the two procedures:

- Pool-then-update gives $Pr_G(A|B) = 0.82$.
- Update-then-pool gives $Pr_G(A) = 0.5$ after learning B.

Constructing the Bets

Three bets, all fair by the group's own lights at the time they're offered.

Bet 1 (before learning whether B): The group buys a conditional bet from the bookie, "if B turns out true, pays \$1 if A", at the group's pooled-prior conditional price of \$0.82. This is a bet that pays \$1 if $A \wedge B$, pays nothing if $A \wedge \neg B$, and the bettor gets their money back (in this case 82 cents) if $\neg B$.

Bet 2 (before learning whether B): The group sells the bookie a small side bet paying \$0.20 if $\neg B$, receiving the fair price of \$0.10.

Bet 3 (after learning B, if B): The group sells a bet paying \$1 if A, receiving the group's updated pooled price of \$0.50.

The Payoff Table

Here's what happens to the group in each possible state of the world:

Table 1: Net payoffs for the group for all three bets in all four world states

	$A \wedge B$	$\neg A \wedge B$	$A \wedge \neg B$	$\neg A \wedge \neg B$
Bet 1	0.18	-0.82	–	–
Bet 2	0.10	0.10	-0.10	-0.10
Bet 3	-0.50	0.50	–	–
Total	-0.22	-0.22	-0.10	-0.10

The group loses money in every cell. That is a sure loss.

In this scenario the group is indifferent between buying and selling each of these bets, but I could have made each bet a penny better for the group (e.g., only asked them to pay 81 cents for the first bet), and the bottom row would still be all negative.

Why This Matters

The bookie isn't cheating. Every bet is offered at a price the group considers fair, given its own credences at the time.

The problem is that the group's pre-experiment conditional price for A given B (0.82) doesn't match the price it actually plans to set after learning B (0.50). This internal inconsistency is exactly what the bookie exploits.

The fundamental problem is the mismatch between Bets 1 and 3, but combined with Bet 2 that leads to sure loss.

Generalizing

The example here is silly; if group members diverge this much, they should talk about it not just pool.

But the problem is general. Whenever group members disagree, there will be some set of bets like this that they will find acceptable, if they use linear pooling, which collectively lead to a sure loss.

In finance terms, it is possible to essentially arbitrage them (though this does take place over time). That's bad!

A Deeper Issue: What Is the Group Learning From?

When individuals update on some evidence E, they're conditioning on E.

But when the pooled credence updates on E, what exactly is the group learning from?

- The group doesn't "observe" E directly.
- The group learns that "all members observed E and updated accordingly."
- These might be subtly different pieces of evidence.

This is why the procedures can diverge.

Why Linear Pooling Fails

The problem is in the structure of linear pooling itself.

When you average probabilities, you're **not** averaging the underlying evidence or likelihood ratios.

- Different people might have different reasons for their credences.
- These reasons interact with new evidence in person-specific ways.
- Averaging the credences loses this information.
- So updating the average is not the same as averaging the updates.

An Intuition

Part of what happens here is that pool-then-update implicitly builds in a judgment of the expertise of the people involved.

When they learn B, that implies 1 was mostly right about A and 2 was mostly wrong. So in future 1 gets more weight.

That doesn't resolve the tension. And it's a big increase in weight for getting one unrelated question right.

What Can We Do About This?

Option 1: Accept the Violation

Maybe commutativity is not as important as we thought.

- Linear pooling has many other nice properties.
- In practice, the violations might be small.
- We could just pick a procedure and stick with it.
- After all, no pooling method satisfies everything.

But this feels unsatisfying—we're essentially admitting the process is arbitrary.

Option 2: Use a Different Pooling Method

Some other pooling methods *do* satisfy external Bayesianity.

In particular: **Geometric pooling** (and more generally, **multiplicative pooling**).

- Instead of averaging probabilities, multiply them.
- More precisely, take weighted geometric means.
- This method commutes with Bayesian updating.

We'll explore this next week.

Option 3: Rethink What We're Pooling

Maybe the problem is that we're pooling the wrong thing.

- Instead of pooling probabilities, pool **likelihood ratios**.
- Or pool **log-odds**.
- Or pool **evidence** rather than opinions. (This is what I think we should do in theory, but it's both hard and impractical.)

Different choices lead to different pooling methods, with different properties.

The Trade-Off

No matter what we do, we face trade-offs.

- Linear pooling is simple and intuitive but violates commutativity.
- Geometric pooling respects commutativity but is less intuitive.
- Other methods have their own strengths and weaknesses.

The right choice depends on what matters most in the context.

Practical Implications

When Does Commutativity Matter Most?

The violation of commutativity is most serious when:

- The group will receive multiple pieces of evidence over time.
- The order in which evidence arrives is variable or arbitrary.
- Procedural manipulation is a real concern.
- The group needs to be seen as rational by external observers.

When Is It Less Important?

Commutativity matters less when:

- The group only needs to form an opinion once, not update repeatedly.
- There's a natural, fixed procedure everyone agrees on in advance.
- Simplicity and transparency are more important than perfect updating.
- The violations in practice are negligibly small.

In democratic decision-making, linear pooling has appeal:

- “One person, one vote” maps naturally to equal weights.
- It’s transparent and easy to explain to citizens.
- The simplicity might outweigh the theoretical problems.

But: For technical committees that update on data over time, commutativity might matter more.

Expert Panels

For scientific advisory committees:

- These groups often update on new studies, data, evidence.
- External Bayesianity ensures consistent, rational updating.
- The committee's reasoning should be defensible.
- This might favor methods that commute with updating.

But: If the committee meets rarely and the updates are large, the method might matter less than the quality of deliberation.

Summing Up

What We've Learned

Linear pooling is:

- Simple and intuitive (weighted averaging).
- Uniquely characterized by unanimity + eventwise independence.
- Also satisfies non-dictatorship and boundedness.
- But violates external Bayesianity (commutativity with updating).

An Impossibility Result

The characterization theorem reveals a fundamental trade-off:

- Linear pooling = Unanimity + Eventwise Independence
- But linear pooling ≠ External Bayesianity
- Therefore: We can't have all three properties together.
- Any pooling method must sacrifice at least one desirable axiom.

This is our first taste of impossibility in opinion pooling.

Violations of external Bayesianity create problems:

- The group's opinion depends on arbitrary procedural choices.
- It enables manipulation through meeting logistics.
- The group can appear irrational to external observers.
- It suggests we're not pooling the right thing.

The Deeper Question

The commutativity problem points to a fundamental issue:

What are we really doing when we pool opinions?

- Are we averaging subjective probabilities?
- Are we combining evidence?
- Are we finding a consensus?
- Are we deferring to a meta-expert?

Different answers lead to different pooling methods.

For Next Time

We'll explore alternatives to linear pooling:

- **Geometric pooling**: Taking geometric means of probabilities.
- **Multiplicative pooling**: Multiplying and normalizing.
- How these methods handle commutativity.
- What other trade-offs they involve.